

Attacking an RSA-like Cryptosystem Using Continued Fractions and Lattices

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Abstract. Let $N = pq$ be the product of two balanced primes. Cota and Teşeleanu (2023) introduced a family of RSA-like cryptosystems defined by $ed - k(p^n - 1)(q^n - 1) = 1$, where $n \geq 1$, encompassing classical RSA ($n = 1$) and the Elkamchouchi–Elshenawy–Shaban variant ($n = 2$). We present a new attack for $n = 3$ that integrates continued fractions with lattice-based methods, naturally extending previous results for $n = 1, 2, 4, 6$.

Keywords: continued fraction attack, lattice attack, small private key attack, RSA

1 Introduction

Background. The RSA cryptosystem, introduced by Rivest, Shamir, and Adleman in 1978 [23], remains one of the most deployed public-key encryption schemes. The textbook RSA works in the multiplicative group $\mathbb{Z}_N^*$, where $N = pq$ is a product of two large primes. Encryption of a message $m \in \mathbb{Z}_N^*$ is performed by $c \equiv m^e \pmod{N}$, with e chosen such that $\gcd(e, \varphi(N)) = 1$, where $\varphi(N) = (p-1)(q-1)$. Decryption is defined as $m \equiv c^d \pmod{N}$, where $d \equiv e^{-1} \pmod{\varphi(N)}$. The tuple (N, e) is public, while (p, q, d) remains secret. In what follows, we restrict attention to balanced primes, meaning $q < p < 2q$, so that p and q share the same bit-length.

From the outset, recovering d from (N, e) has been a central target for cryptanalysis. Wiener’s classical result [29] shows that if $d < N^{0.25}/3$, it can be recovered from the continued fraction expansion of e/N , which in turn factors N . Boneh and Durfee [4] improved this to $d < N^{0.292}$ via Coppersmith’s method [8] and lattice reduction [17], with Herrmann and May [14] later achieving the same bound using simpler tools. Broader surveys of such attacks are given in [3, 19, 24].

RSA over Gaussian Integers. In 2002, Elkamchouchi, Elshenawy, and Shaban [12] proposed an RSA analogue over the ring of Gaussian integers modulo N . Elements have the form $a + bi$ with $a, b \in \mathbb{Z}_N$ and $i^2 = -1$. The multiplicative group $\mathbb{Z}_N[i]$ has order $\phi(N) = (p^2 - 1)(q^2 - 1)$, and the exponents satisfy $\gcd(e, \phi(N)) = 1$ with $d \equiv e^{-1} \pmod{\phi(N)}$. Encryption and decryption proceed exactly as in RSA, except all arithmetic is carried out in $\mathbb{Z}_N[i]$.

Although this extension was claimed to offer greater security, Bunder *et al.* [5] showed a Wiener-type attack via continued fractions. Later improvements [22,31] using lattice reduction techniques pushed the bound to $d < N^{0.585}$. Additional analysis can be found in [10,24].

Generalizing via Galois Fields. The rings Z_p and $Z_p[i]$ can be identified as $Z_p \cong GF(p)$ and $Z_p[i] \cong GF(p^2)$, where GF denotes a Galois field. Thus, classical RSA operates over $GF(p) \times GF(q)$, while the Gaussian variant corresponds to $GF(p^2) \times GF(q^2)$. This perspective led Cotan and Teşeleanu [10] to define a family of RSA-like systems over $GF(p^n) \times GF(q^n)$, for $n \geq 1$, with group order $\varphi_n(N) = (p^n - 1)(q^n - 1)$. Encryption and decryption generalize directly from the $n = 1, 2$ cases.

The main motivation was to determine whether Wiener-type cryptanalysis extends to this broader class. Indeed, [10] showed that for $d < N^{0.25n}$, continued fractions recover d for any n . The unbalanced prime case was later covered in [11]. A lattice-based extension, left as an open question in these works, was resolved in [26], yielding stronger bounds. Other related lattice techniques appear in [28].

Related work. The approach of Blomer and May [1] combined continued fractions with Coppersmith’s method. They showed that if

$$ae + b = k\varphi_1(N),$$

and

$$0 < a \leq \frac{1}{3}N^{\frac{1}{4}} \quad \text{and} \quad |b| = \mathcal{O}(N^{-\frac{3}{4}}ae),$$

then we can factor N in polynomial time.

Another variation, due to Nitaj [21], considered

$$ae - b(p - u)(q - u) = 1,$$

under specific bounds on a, b, u, v and with small prime factors for $p - u$ and $q - v$.

The $n = 2, 4, 6$ cases were analyzed in [6,20,27], where factorization follows if a, b and c from $ae - b\varphi_n(N) = c$ satisfy specific conditions. We summarize these results in Table 1.

Our Contributions. We investigate the previously untreated case $n = 3$, by combining continued fractions with lattice reduction techniques. Our method begins by recovering integers a, b from $ae - b\varphi_3(N) = c$ via a continued fraction approximation. These values yield an estimate $\hat{p}$ for p , which can then be recovered exactly using Coppersmith’s method. Knowing the upper bound on d from $ed - k\varphi_3(N) = 1$ also allows us, through [20], to deduce a corresponding lower bound.

	Upper bound
$n = 2$ [6]	$2N - 4\sqrt{2}N^{3/4}$
$n = 3$ (this work)	$\frac{N^3 - 10N\sqrt{N} + 1}{22N\sqrt{N} + N^{3/4}}$
$n = 4$ [20]	$\frac{2N^4 - 49N^2 + 2}{170N^2 + 4N}$
$n = 6$ [27]	$\frac{N^6 - 162N^3 + 1}{1100N^3 + 2N\sqrt{N}}$

Table 1. Summary on upper bounds for ab .

Structure of the Paper. Preliminaries are reviewed in Section 2. The attack is detailed in Section 3, followed by an example in Section 4 and final remarks in Section 5.

2 Preliminaries

Notations. Throughout the paper, λ denotes a security parameter. Also, the notation $|S|$ denotes the cardinality of a set S . The action of selecting a random element x from a sample space X is denoted by $x \xleftarrow{\$} X$.

2.1 Continued fraction

For any real number ζ there exists a unique sequence $(a_n)_n$ of integers such that

$$\zeta = a_0 + \frac{1}{a_1 + \frac{1}{a_2 + \frac{1}{a_3 + \frac{1}{a_4 + \dots}}}},$$

where $a_k > 0$ for any $k \geq 1$. This sequence represents the continued fraction expansion of ζ and is denoted by $\zeta = [a_0, a_1, a_2, \dots]$. Remark that ζ is a rational number if and only if its corresponding representation as a continued fraction is finite.

For any real number $\zeta = [a_0, a_1, a_2, \dots]$, the sequence of rational numbers $(A_n)_n$, obtained by truncating this continued fraction, $A_k = [a_0, a_1, a_2, \dots, a_k]$, is called the convergents sequence of ζ .

According to [13], the following bound allows us to check if a rational number u/v is a convergent of ζ .

Theorem 1. *Let $\zeta = [a_0, a_1, a_2, \dots]$ be a positive real number. If u, v are positive integers such that $\gcd(u, v) = 1$ and*

$$\left| \zeta - \frac{u}{v} \right| < \frac{1}{2v^2},$$

then u/v is a convergent of $[a_0, a_1, a_2, \dots]$.

2.2 Finding Small Roots

In this section, we outline some tools used for solving the problem of finding small roots, both in the modular and integer cases.

Coppersmith [7–9] provided rigorous techniques for computing small integer roots of single-variable polynomials modulo an integer, as well as bivariate polynomials over the integers. In the case of modular roots, Coppersmith’s ideas were reinterpreted by Howgrave-Graham [15]. We further provide Howgrave-Graham result.

Theorem 2. *Let $f(x_1, \dots, x_n) = \sum a_{i_1 \dots i_n} x_1^{i_1} \dots x_n^{i_n} \in \mathbb{Z}[x_1, \dots, x_n]$ be a polynomial with at most ω monomials, α be an integer and let*

$$\|f(x_1, \dots, x_n)\| = \sqrt{\sum |a_{i_1 \dots i_n}|^2}$$

be its norm. Suppose that

- $f(y_1, \dots, y_n) \equiv 0 \pmod{\alpha}$ for some $|y_1| < X_1, \dots, |y_n| < X_n$,
- $\|f(y_1 X_1, \dots, y_n X_n)\| < \alpha/\sqrt{\omega}$,

then $f(y_1, \dots, y_n) = 0$ holds over integers.

Lenstra, Lenstra and Lovász [17] proposed a lattice reduction algorithm (LLL) that is widely used in cryptanalysis and is typically combined with Howgrave-Graham’s lemma. We further provide the version presented in [16, 18].

Theorem 3. *Let L be a lattice of dimension ω . In polynomial time, the LLL algorithm outputs a reduced basis $(b_1, \dots, b_\omega)$ that satisfies*

$$\|b_1\| \leq \dots \leq \|b_i\| \leq 2^{\frac{\omega(\omega-1)}{4(\omega+1-i)}} \det(L)^{\frac{1}{\omega+1-i}},$$

where $\det(L)$ is the determinant of lattice L .

Note that the condition

$$2^{\frac{\omega(\omega-1)}{4(\omega+1-i)}} \det(L)^{\frac{1}{\omega+1-i}} < \alpha/\sqrt{\omega}$$

implies that the polynomials corresponding to b_i match Howgrave-Graham’s bound. This leads to

$$\det(L) \leq \varepsilon \alpha^{\omega+1-i},$$

where ε is an error term that is usually ignored.

In order to find a solution $(y_1, \dots, y_n)$ we need the following assumption to be true.

Assumption 4 *The LLL reduced basis polynomials are algebraically independent¹, and the resultant computations for b_i yield the common roots of these polynomials.*

In [2], the authors present a more flexible formulation of Coppersmith's result [8]. Their method first constructs a specific lattice basis, applies the LLL algorithm [17] to reduce it, and finally uses Howgrave-Graham's lemma [15] to derive the solutions.

Theorem 5. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Suppose we are given an approximation of p with additive error at most $N^{1/4}$. Then N can be factored in polynomial time.*

Once an attack is obtained for a given upper bound on small private exponents, the result of [20] implies the existence of a corresponding attack for a given lower bound on large private exponents. For results concerning $\varphi_1, \varphi_2, \varphi_4$ and φ_6 , we refer the reader to [20, 27].

Theorem 6. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Let $\psi : \mathbb{N} \times \mathbb{N} \rightarrow \mathbb{N}$. Suppose we are given an algorithm $\mathcal{A}$ that is able to factor N in polynomial time given N and a public exponent $0 < e < \psi(p, q)$ such that there exists positive integers x and y satisfying*

$$ex - y\phi(p, q) = z, \text{ with } xy < \mathcal{B}_1 \text{ and } |z| < \mathcal{B}_2,$$

for $\mathcal{B}_1 > 0$ and $\mathcal{B}_2 \geq 1$. Then, using algorithm $\mathcal{A}$, N can be factored in polynomial time given N and a public exponent $0 < e' < \psi(p, q)$ such that the corresponding private exponent is $d' = \psi(p, q) - d$ for some $d < \sqrt{\mathcal{B}_1}$.

2.3 Quotient Groups

In this section we provide the group theory needed to introduce the RSA-like family. Therefore, let $(\mathbb{F}, +, \cdot)$ be a field and $t^n - r$ an irreducible polynomial in $\mathbb{F}[t]$. Then

$$\mathbb{A}_n = \mathbb{F}[t]/(t^n - r) = \{a_0 + a_1t + \dots + a_{n-1}t^{n-1} \mid a_0, a_1, \dots, a_{n-1} \in \mathbb{F}\}$$

is the corresponding quotient field. Let $a(t), b(t) \in \mathbb{A}_n$. Remark that the quotient field induces a natural product

$$\begin{aligned} a(t) \circ b(t) &= \left(\sum_{i=0}^{n-1} a_i t^i \right) \circ \left(\sum_{j=0}^{n-1} b_j t^j \right) \\ &= \sum_{i=0}^{n-2} \left(\sum_{j=0}^i a_j b_{i-j} + r \sum_{j=0}^{i+n} a_j b_{i-j+n} \right) t^i + \sum_{j=0}^{n-1} a_j b_{n-1-j} t^{n-1}. \end{aligned}$$

¹ They do not share a non-trivial gcd.

2.4 RSA-like Cryptosystems

Let p be a prime number. When we instantiate $\mathbb{F} = \mathbb{Z}_p$, we have that $\mathbb{A}_n = GF(p^n)$ is the Galois field of order p^n . Moreover, $\mathbb{A}_n^*$ is a cyclic group of order $\varphi_n(\mathbb{Z}_p) = p^n - 1$. Remark that an analogous of Fermat's little theorem holds

$$a(t)^{\varphi_n(\mathbb{Z}_p)} \equiv 1 \pmod{p},$$

where $a(t) \in \mathbb{A}_n^*$ and the power is evaluated by $\circ$ -multiplying $a(t)$ by itself $\varphi_n(\mathbb{Z}_p) - 1$ times. Based on these observations, the authors of [10] built an encryption scheme that is similar to RSA by using the $\circ$ operation as the product.

Setup(λ): Let $n \geq 1$ be an integer. Randomly generate two distinct large prime numbers p, q such that $p, q \geq 2^\lambda$ and compute their product $N = pq$. Select $r \in \mathbb{Z}_N$ such that the polynomial $t^n - r$ is irreducible in $\mathbb{Z}_p[t]$ and $\mathbb{Z}_q[t]$. Let

$$\varphi_n(\mathbb{Z}_N) = \varphi_n(N) = (p^n - 1) \cdot (q^n - 1).$$

Choose an integer e such that $\gcd(e, \varphi_n(N)) = 1$ and compute d such that $ed \equiv 1 \pmod{\varphi_n(N)}$. Output the public key $pk = (n, N, r, e)$. The corresponding secret key is $sk = (p, q, d)$.

Encrypt(pk, m): To encrypt a message $m = (m_0, \dots, m_{n-1}) \in \mathbb{Z}_N^n$ we first construct the polynomial $m(t) = m_0 + \dots + m_{n-1}t^{n-1} \in \mathbb{A}_n^*$ and then we compute $c(t) \equiv [m(t)]^e \pmod{N}$. Output the ciphertext $c(t)$.

Decrypt($sk, c(t)$): To recover the message, simply compute $m(t) \equiv [c(t)]^d \pmod{N}$ and reassemble $m = (m_0, \dots, m_{n-1})$.

Remark 1. When $n = 1$ we get the RSA scheme [23]. Also, when $n = 2$, we obtain the ElGamal cryptosystem [12].

3 A Generalized Wiener-type Attack

In this section, we investigate the generalized equation $ae - b\varphi_3(N) = c$, where e is the public exponent and c is a known value. Our approach unfolds in two main stages: initially, we derive the coefficients a and b through a continued fraction expansion; subsequently, we approximate p , and use Coppersmith's result to factor N . We start by examining the lattice-based approach and then proceed to the continued fractions method.

3.1 Application of Lattices

We begin this subsection with a lemma that provides bounds for the sum $p^3 + q^3$.

Lemma 1. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Then the following property holds*

$$2N\sqrt{N} < p^3 + q^3 < \frac{9\sqrt{2}}{4}N\sqrt{N}.$$

Proof. From the inequality $q < p < 2q$ we derive $1 < p\sqrt{p}/(q\sqrt{q}) < 2\sqrt{2}$. Since the function $f(x) = x + 1/x$ is increasing on $[1, +\infty)$, we have that

$$2 < \frac{p\sqrt{p}}{q\sqrt{q}} + \frac{q\sqrt{q}}{p\sqrt{p}} < \frac{9\sqrt{2}}{4}.$$

Multiplying the inequality with $N\sqrt{N}$, we obtain

$$2N\sqrt{N} < p^3 + q^3 < \frac{9\sqrt{2}}{4}N\sqrt{N}.$$

just as desired. $\square$

Using the following lemma (provided in [21, Lemma 1]), we prove that $p - q = \Omega(N\sqrt{N})$.

Lemma 2. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Then the following property holds*

$$\frac{\sqrt{2}}{2}\sqrt{N} < q < \sqrt{N} < p < \sqrt{2}\sqrt{N}.$$

Lemma 3. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Then the following property holds*

$$0 < p^3 - q^3 < \frac{7\sqrt{2}}{4}N\sqrt{N}.$$

Proof. According to Lemma 2 we obtain

$$\frac{\sqrt{2}}{4}N\sqrt{N} < q^3 < N\sqrt{N} < p^3 < 2\sqrt{2}N\sqrt{N}.$$

Therefore, we have

$$0 < p^3 - q^3 < 2\sqrt{2}N\sqrt{N} - \frac{\sqrt{2}}{4}N\sqrt{N}.$$

just as desired. $\square$

Let $S_3 = p^3 + q^3$ and $D_3 = p^3 - q^3$. Using the value for φ_3 , namely

$$\varphi_3 = N^3 - S_3 + 1,$$

and the relation

$$p^3 - q^3 = \sqrt{(p^3 + q^3)^2 - 4N^3}$$

we further derive some approximations for $p^3 + q^3$ and $p^3 - q^3$.

Lemma 4. *Let $0 < \delta < 7\sqrt{2}/4$ and $N = pq > (3\sqrt{2}/4\delta)^{4/3}$ be the product of two unknown primes with $q < p < 2q$ and $p^3 - q^3 \geq \delta N\sqrt{N}$. Also, let e be a public exponent satisfying $ae - b\varphi_3(N) = c$ such that $6|c| < \delta bN^{3/4}$. We define $S_3 = p^3 + q^3$, $D_3 = p^3 - q^3$,*

$$\hat{S}_3 = N^3 - 1 - \frac{ae}{b} \quad \text{and} \quad \hat{D}_3 = \sqrt{\hat{S}_3^2 - 4N^3}$$

Then the following hold

$$|S_3 - \hat{S}_3| < \frac{3}{6}N^{3/4} \quad \text{and} \quad |D_3 - \hat{D}_3| < \frac{9}{6}N^{3/4}.$$

Proof. We know that

$$\varphi_3 = N^3 - p^3 - q^3 + 1 = (ae - c)/b,$$

and thus, we have that

$$S_3 = N^3 + 1 - \frac{ae - c}{b}.$$

Therefore, we obtain the following

$$|S_3 - \hat{S}_3| = \frac{|c|}{b} < \frac{\delta}{6}N^{3/4} < \frac{3}{6}N^{3/4}.$$

For the second part of the proof, we first observe that

$$\hat{S}_3^2 - 4N^3 = D_3^2 - 2S_3\frac{c}{b} + \frac{c^2}{b^2}.$$

To prove that $\hat{D}_3$ is well defined, it suffices to show that $D_3^2 \geq 2S_3\frac{|c|}{b}$. We observe that

$$2S_3\frac{|c|}{b} < 2 \cdot \frac{9\sqrt{2}}{4}N\sqrt{N} \cdot \frac{\delta}{6}N^{3/4} = \frac{3\sqrt{2}}{4}\delta N^{9/4} < \delta^2 N^3 < (p^3 - q^3)^2.$$

Note that $\delta N^{3/4}/6 < N\sqrt{N}/2 < (p^3 + q^3)/4$ and thus $|\hat{S}_3| < 5S_3/4$. Using

$$\hat{D}_3 - D_3 = \sqrt{\hat{S}_3^2 - 4N^3} - D_3 = \frac{(\hat{S}_3 - S_3)(\hat{S}_3 + S_3)}{\hat{D}_3 + D_3}$$

we obtain that

$$\hat{D}_3 - D_3 < \frac{\delta}{6}N^{3/4} \cdot \frac{9}{4} \cdot 4N\sqrt{N} \cdot \frac{1}{D_3} < \frac{9}{6}N^{3/4},$$

just as desired. □

The following lemma proven in [27] will be useful to prove the subsequent theorem.

Lemma 5. *Let $u > v > 0$. The following inequality holds*

$$\sqrt[3]{u} - \sqrt[3]{v} < \sqrt[3]{u \pm v} < \sqrt[3]{u} + \sqrt[3]{v}.$$

We are now in a position to apply Coppersmith's result to factor N .

Theorem 7. *Let $0 < \delta < 7\sqrt{2}/4$ and $N = pq > (3\sqrt{2}/4\delta)^{4/3}$ be the product of two unknown primes with $q < p < 2q$ and $p^3 - q^3 \geq \delta N\sqrt{N}$. Also, let e be a public exponent satisfying $ae - b\varphi_3(N) = c$ such that $6|c| < b\delta N^{3/4}$. Given e , N , a and b we can factor N in polynomial time.*

Proof. Using the approximations derived in Lemma 4 we have that

$$\begin{aligned} \left| p - \sqrt[3]{\frac{1}{2}(\hat{S}_3 + \hat{D}_3)} \right| &= \left| \sqrt[3]{\frac{1}{2}(S_3 + D_3)} - \sqrt[3]{\frac{1}{2}(\hat{S}_3 + \hat{D}_3)} \right| \\ &\leq \sqrt[3]{\frac{1}{2}|S_3 - \hat{S}_3| + \frac{1}{2}|D_3 - \hat{D}_3|} \\ &< \sqrt[3]{\frac{3}{12}N^{3/4} + \frac{9}{12}N^{3/4}} = N^{1/4}, \end{aligned}$$

where for the first inequality we used Lemma 5. Therefore,

$$\hat{p} = \sqrt[3]{0.5(\hat{S} + \hat{D})}$$

is a good approximation of p . Now according to Theorem 5, we can factor N in polynomial time. $\square$

3.2 Application of Continued Fractions

We begin this subsection with a lemma (provided in [6, Lemma 3]) that provides lower and upper bounds for p and q .

Lemma 6. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Then the following property holds*

$$2\sqrt{N} < p + q < \frac{3\sqrt{2}}{2}\sqrt{N}.$$

We further derive an useful bound for the continued fraction part of our attack.

Lemma 7. *Let $N = pq$ be the product of two unknown primes with $q < p < 2q$. Then the following property holds*

$$\left| S^3 - 3NS - 10N\sqrt{N} \right| < 11N\sqrt{N}$$

where $S = p + q$.

Proof. Using Lemma 6 we obtain that

$$8N\sqrt{N} < S^3 < 27N\sqrt{N}.$$

Therefore, we have

$$-N\sqrt{N} = (8 - 3 \cdot 3)N\sqrt{N} < S^3 - 3NS < (27 - 3 \cdot 2)N\sqrt{N} = 21N\sqrt{N}.$$

Thus, we obtain

$$-11N\sqrt{N} < A - 10N\sqrt{N} < 11N\sqrt{N},$$

just as desired. $\square$

Theorem 8. *Let $0 < \delta < 7\sqrt{2}/4$ and $N = pq > (3\sqrt{2}/4\delta)^{4/3}$ be the product of two unknown primes with $q < p < 2q$ and $p^3 - q^3 \geq \delta N\sqrt{N}$. Also, let e be a public exponent satisfying $ae - b\varphi_3(N) = c$ such that $6|c| < \delta bN^{3/4}$. Given e , N we can factor N in polynomial time if*

$$ab < \frac{N^3 - 10N\sqrt{N} + 1}{N^{3/4} + 22N\sqrt{N}}.$$

Proof. According to [26] we have

$$\varphi_3 = N^3 + 3NS - S^3 + 1.$$

We denote by

$$A = S^3 - 3NS - 10N\sqrt{N}.$$

We know that

$$ae - b(N^3 - A - 10N\sqrt{N} + 1) = c$$

which is equivalent to

$$ae - b(N^3 - 10N\sqrt{N} + 1) = c - bA.$$

Dividing everything by $a(N^3 - 10N\sqrt{N} + 1)$ we obtain

$$\frac{e}{N^3 - 10N\sqrt{N} + 1} - \frac{b}{a} = \frac{c - bA}{a(N^3 - 10N\sqrt{N} + 1)}.$$

Taking the absolute value we obtain

$$\begin{aligned} \left| \frac{e}{N^3 - 10N\sqrt{N} + 1} - \frac{b}{a} \right| &\leq \frac{|c| + |bA|}{a(N^3 - 10N\sqrt{N} + 1)} \\ &\leq \frac{\delta N^{3/4} + 6|A|}{6(N^3 - 10N\sqrt{N} + 1)} \cdot \frac{b}{a} \\ &\leq \frac{N^{3/4} + 22N\sqrt{N}}{N^3 - 10N\sqrt{N} + 1} \cdot \frac{b}{2a} \\ &\leq \frac{1}{ab} \cdot \frac{b}{2a} = \frac{1}{2a^2}. \end{aligned}$$

where for the third inequality we used Lemma 7 and for the last inequality we used our hypothesis. Since

$$\left| \frac{e}{N^3 - 10N\sqrt{N} + 1} - \frac{b}{a} \right| \leq \frac{1}{2a^2}.$$

then according to Theorem 1 b/a appears among the convergents of $e/(N^3 - 10N\sqrt{N} + 1)$. Once we obtain a and b we apply Theorem 7, and thus we conclude our proof. $\square$

To conclude, we apply our general result to the RSA-like cryptosystem in the case $n = 3$. The second corollary follows from Theorem 6.

Corollary 1. *Let $0 < \delta < 7\sqrt{2}/4$ and $N = pq > (3\sqrt{2}/4\delta)^{4/3}$ be the product of two unknown primes with $q < p < 2q$ and $p^3 - q^3 \geq \delta N\sqrt{N}$. Also, let $e < \varphi_3(N)$ be a public exponent satisfying $ed - k\varphi_3(N) = 1$. Given e, N we can factor N in polynomial time if*

$$d < \sqrt{\frac{N^3 - 10N\sqrt{N} + 1}{N^{3/4} + 22N\sqrt{N}}}.$$

Proof. We notice that

$$k = \frac{ed - 1}{\varphi_3(N)} < \frac{ed}{\varphi_3(N)} < d$$

and

$$kd < d^2 < \frac{N^3 - 10N\sqrt{N} + 1}{N^{3/4} + 22N\sqrt{N}}$$

According to Theorem 8 we can factor N in polynomial time. $\square$

Corollary 2. *Let $0 < \delta < 7\sqrt{2}/4$ and $N = pq > (3\sqrt{2}/4\delta)^{4/3}$ be the product of two unknown primes with $q < p < 2q$ and $p^3 - q^3 \geq \delta N\sqrt{N}$. Also, let $e < \varphi_3(N)$ be a public exponent satisfying $ed - k\varphi_3(N) = 1$. Given e, N we can factor N in polynomial time if*

$$d > \varphi_3(N) - \sqrt{\frac{N^3 - 10N\sqrt{N} + 1}{N^{3/4} + 22N\sqrt{N}}}.$$

4 Experimental Results

To validate our result, we executed the code for our attack [25] on a workstation running Ubuntu 20.04.1, equipped with an Intel(R) Core(TM) i7-1165G7 CPU at 2.80 GHz (8 cores) and 16 GB of RAM. The implementation was carried out in SageMath 10.3, based on the Coppersmith attack code from [30].

We used the following parameters

$$\begin{aligned} N &= 3489655588599196597998727781564283681960038261038493763731, \\ e &= 91438868789576831711580148107321175513335455073330425214416055 \\ &\quad 247322855089745802781880650112500450644009507380804851765908646 \\ &\quad 4747764892583247102010073965492533357516309277. \end{aligned}$$

Computing the continued fraction expansion of $e/(N^3 - 10N\sqrt{N} + 1)$, we get the first 25 partial quotients

$$[0, 4, 1, 1, 1, 5, 8, 5, 2, 12, 2, 5, 1, 2, 1, 12, 42, 10, 2, 1, 1, 9, 1, 2, 1, \dots].$$

Looking at the 82th convergent we obtain

$$\begin{aligned} a &= 1393796574908163946345982392040522594123813 \\ b &= 299904193479875310237724396865170476455978, \end{aligned}$$

which satisfy the condition of Theorem 8. Therefore, we obtain the following approximations of S_3 and D_3

$$\begin{aligned} \hat{S}_3 &= 233338672780168639333586294670548066997335251906712757949 \\ &\quad 9445675397277719907479079497489 \\ \hat{D}_3 &= 229667362942519266201139773721266646133097156290727851611 \\ &\quad 6047864086018875551724069122546. \end{aligned} \tag{1}$$

Once we know $\hat{S}_3$ and $\hat{D}_3$ we can compute p 's approximation

$$\hat{p} = 132287523294776463864922187740.$$

Using Coppersmith's algorithm we find

$$p = 132287523294776463864922187741,$$

and then we can compute

$$q = N/p = 26379325137285943549540377391.$$

5 Conclusions

In this paper, we have presented a generalized Wiener-type attack against RSA-like cryptosystems. Our approach begins by analyzing the general equation $ae - b\varphi_3(N) = c$, followed by the application of a result due to Coppersmith [2, 8]. We demonstrate that when d is either sufficiently small or sufficiently large, N can be factored in polynomial time.

Future work. An interesting direction for future work would be to develop a method applicable to $\varphi_i(N)$ for arbitrary i , rather than only for specific cases (*i.e.* for $n = 1, 2, 3, 4, 6$).

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